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-- SECTION D : Joins Queries

-- Database : Celebal_Week2

-- Table : dbo.Superstore (single flat table — self-join

-- and derived-table joins used to simulate

-- multi-table relational analysis)

-- Tool : SQL Server Management Studio (SSMS)

-- Author : ARATI THORAT | Celebal Internship – Week 2

-- =====

USE Celebal_Week2;

GO

-- NOTE

-- Since the Superstore CSV was imported as a single flat table,

-- we use SELF-JOINS and SUBQUERY-based derived tables to

-- demonstrate INNER JOIN and LEFT JOIN logic meaningfully.

-- This mirrors real-world scenarios where one large table

-- is compared against itself or against aggregated views.

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-- D1. INNER JOIN Simulation

-- PURPOSE : Join customer orders with their regional average

-- to compare individual orders vs regional baseline

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SELECT

    s.Order_ID,

    s.Customer_Name,

    s.Region,

    s.Category,

    s.Product_Name,

    ROUND(s.Sales, 2) AS Order_Sales,

    ROUND(s.Profit, 2) AS Order_Profit,

    ROUND(ra.Region_Avg, 2) AS Regional_Avg_Sales,

    CASE

        WHEN s.Sales > ra.Region_Avg

        THEN 'Above Average'

        ELSE 'Below Average'

    END AS Sales_vs_Region

FROM dbo.Superstore AS s

INNER JOIN (

    -- Derived table: average sales per region

    SELECT Region, AVG(Sales) AS Region_Avg

    FROM   dbo.Superstore

    GROUP BY Region

) AS ra ON s.Region = ra.Region

ORDER BY s.Region, s.Sales DESC;

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Order_ID	Customer_Name	Region	Category	Product_Name	Order_Sales	Order_Profit	Regional_Avg_Sales	Sales_vs_Region
1	CA-2016-118689	Tamara Chand	Central	Technology	Canon imageCLASS 2200 Advanced Copier	17499.95	8399.98	215.77
2	CA-2016-111721	Adrian Barton	Central	Office Supplies	GBC Ibmaster 500 Manual ProClick Binding System	9892.74	4946.37	215.77
3	CA-2014-116904	Sanjit Chand	Central	Office Supplies	Ibico EPK-21 Electric Binding System	9449.95	4630.48	215.77
4	CA-2014-139892	Becky Martin	Central	Technology	Lexmark MX611dhe Monochrome Laser Printer	8159.95	-1359.99	215.77
5	CA-2017-136289	Andy Reiter	Central	Office Supplies	GBC DocuBind P400 Electric Binding System	5443.96	2504.22	215.77
6	US-2015-126587	Harry Marie	Central	Technology	Canon PC1060 Personal Laser Copier	4899.93	2302.97	215.77
7	US-2017-167402	Cathy Prescott	Central	Office Supplies	Martin Vale Chadless Opener Electric Letter Opener	4164.05	83.28	215.77
8	CA-2016-103982	Alex Avila	Central	Office Supplies	High Speed Automatic Electric Letter Opener	3930.07	-786.01	215.77
9	CA-2015-120782	Shirley Daniels	Central	Office Supplies	Fellowes PB500 Electric Punch Plastic Comb Bind...	3812.97	1906.48	215.77
10	CA-2016-122903	Laura Armstrong	Central	Furniture	HON 5400 Series Task Chairs for Big and Tall	3504.9	700.98	215.77
11	US-2014-106992	Sean Braxton	Central	Technology	Lexmark MX611dhe Monochrome Laser Printer	3059.98	-510	215.77
12	CA-2017-159366	Bart Watters	Central	Technology	Lexmark MX611dhe Monochrome Laser Printer	3059.98	680	215.77
13	CA-2014-120474	Resi Polking	Central	Furniture	Office Star - Professional Matrix Back Chair with 2-to...	2807.84	673.88	215.77
14	US-2015-100377	Todd Sumrall	Central	Technology	Canon Imageclass D680 Copier / Fax	2799.96	874.99	215.77
15	CA-2014-154627	Sue Ann Reed	Central	Technology	Apple iPhone 5S	2735.95	341.99	215.77
16	US-2016-164196	Alejandro Savely	Central	Furniture	Balt Solid Wood Round Tables	2678.94	241.1	215.77
17	CA-2015-120677	Bill Donatelli	Central	Furniture	Hon Pagoda Stacking Chairs	2567.84	770.35	215.77
18	US-2014-106992	Sean Braxton	Central	Technology	Xerox WorkCentre 6505DN Laser Multifunction Prin...	2519.96	-252	215.77
19	CA-2017-160885	Juliana Krohn	Central	Technology	ClearOne CHATAttach 160 - speaker phone	2479.96	743.99	215.77
20	CA-2015-139731	Joel Eaton	Central	Furniture	HON 5400 Series Task Chairs for Big and Tall	2453.43	-350.49	215.77
21	CA-2017-121559	Helen Wass...	Central	Office Supplies	Honeywell Enviracaire Portable HEPA Air Cleaner fo...	2405.2	793.72	215.77
22	US-2017-141677	Heather Kirkland	Central	Technology	Canon PC1080F Personal Copier	2399.96	569.99	215.77
23	CA-2016-108987	Anna Gayman	Central	Furniture	Riverside Palais Royal Lawyers Bookcase, Royale C...	2396.27	-317.15	215.77
24	CA-2014-169019	Luke Foster	Central	Office Supplies	GBC DocuBind P400 Electric Binding System	2177.58	-3701.89	215.77
25	CA-2017-121790	Liz Preis	Central	Technology	Samsung Galaxy S4	2003.17	250.4	215.77
26	CA-2014-131926	Dianna Wilson	Central	Furniture	Global Deluxe High-Back Manager's Chair	2001.86	580.54	215.77
27	CA-2014-163748	Hunter Glantz	Central	Technology	Hewlett Packard 610 Color Digital Copier / Printer	1999.96	624.99	215.77

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OBSERVATION:

West region has the highest regional average sale.

Most high-value Technology orders are 'Above Average' within their region — confirming Tech is the growth driver everywhere.

-- D2. INNER JOIN – Order Detail with Category Average

-- PURPOSE : Find orders that exceed the category average sale

SELECT

s.Order_ID,

s.Order_Date,

s.Customer_Name,

s.Category,

s.Product_Name,

ROUND(s.Sales, 2) AS Order_Sales,

ROUND(ca.Cat_Avg, 2) AS Category_Avg_Sales,

ROUND(s.Profit, 2) AS Profit

FROM dbo.Superstore AS s

INNER JOIN (

SELECT Category, AVG(Sales) AS Cat_Avg

FROM dbo.Superstore

GROUP BY Category

) AS ca ON s.Category = ca.Category

WHERE s.Sales > ca.Cat_Avg

ORDER BY s.Category, s.Sales DESC;

Results		Messages							
	Order_ID	Order_Date	Customer_Name	Category	Product_Name	Order_Sales	Category_Avg_Sales	Profit	
1	CA-2017-118892	2017-08-17	Tom Prescott	Furniture	HON 5400 Series Task Chairs for Big and Tall	4416.17	349.83	-630.88	
2	CA-2015-117086	2015-11-08	Quincy Jones	Furniture	Riverside Palais Royal Lawyers Bookcase, Royale Ch...	4404.9	349.83	1013.13	
3	CA-2015-116638	2015-01-28	Joseph Holt	Furniture	Chromcraft Bull-Nose Wood Oval Conference Tables ...	4297.64	349.83	-1862.31	
4	US-2015-126977	2015-09-17	Peter Fuller	Furniture	Riverside Palais Royal Lawyers Bookcase, Royale Ch...	4228.7	349.83	158.58	
5	CA-2014-128209	2014-11-17	Greg Tran	Furniture	DMI Eclipse Executive Suite Bookcases	4007.84	349.83	-50.1	
6	CA-2014-116246	2014-09-12	Luke Weiss	Furniture	HON 5400 Series Task Chairs for Big and Tall	3785.29	349.83	420.59	
7	CA-2014-168494	2014-12-12	Nora Preis	Furniture	Bretford Rectangular Conference Table Tops	3610.85	349.83	135.41	
8	CA-2016-122903	2016-05-27	Laura Armstrong	Furniture	HON 5400 Series Task Chairs for Big and Tall	3504.9	349.83	700.98	
9	CA-2016-107104	2016-11-26	Maribeth Schnelling	Furniture	DMI Eclipse Executive Suite Bookcases	3406.66	349.83	160.31	
10	CA-2015-164301	2015-03-26	Ellis Ballard	Furniture	Bush Advantage Collection Racetrack Conference Ta...	3393.68	349.83	610.86	
11	US-2015-150630	2015-09-17	Tracy Blumstein	Furniture	Riverside Palais Royal Lawyers Bookcase, Royale Ch...	3083.43	349.83	-1665.05	
12	CA-2017-100111	2017-09-20	Seth Vernon	Furniture	Hon 4070 Series Pagoda Armless Upholstered Stack...	2888.13	349.83	609.72	
13	US-2016-126452	2016-08-21	Scott Coram	Furniture	Bretford CR8500 Series Meeting Room Furniture	2887.06	349.83	180.44	
14	CA-2014-120474	2014-12-01	Resi Polking	Furniture	Office Star - Professional Matrix Back Chair with 24p...	2807.84	349.83	673.88	
15	CA-2017-102204	2017-05-01	Caroline Jumper	Furniture	HON 5400 Series Task Chairs for Big and Tall	2803.92	349.83	0	
16	CA-2015-164777	2015-01-27	Sean Christensen	Furniture	HON 5400 Series Task Chairs for Big and Tall	2803.92	349.83	0	
17	US-2016-164196	2016-11-11	Alejandro Savely	Furniture	Balt Solid Wood Round Tables	2678.94	349.83	241.1	
18	CA-2015-111038	2015-12-01	Lindsay Castell	Furniture	GuestStacker Chair with Chrome Finish Legs	2676.67	349.83	267.67	
19	CA-2017-123967	2017-11-01	Sarah Foster	Furniture	Hon 94000 Series Round Tables	2665.62	349.83	239.91	
20	CA-2015-160227	2015-11-02	Emily Ducich	Furniture	Hon Olson Stacker Chairs	2621.32	349.83	553.39	
21	CA-2014-167199	2014-01-06	Maria Elezadi	Furniture	Global Deluxe High-Back Manager's Chair	2573.82	349.83	746.41	
22	CA-2015-120677	2015-05-31	Bill Donatelli	Furniture	Hon Pagoda Stacking Chairs	2567.84	349.83	770.35	
23	CA-2016-169670	2016-12-25	Joe Elijah	Furniture	Hon 4700 Series Mobius Mid-Back Task Chairs with ...	2563.06	349.83	313.26	
24	CA-2015-139731	2015-10-15	Joel Eaton	Furniture	HON 5400 Series Task Chairs for Big and Tall	2453.43	349.83	-350.49	
25	CA-2016-104157	2016-07-25	Meg Tillman	Furniture	Office Impressions End Table, 20-1/2"H x 24"W x 20"D	2430.08	349.83	388.81	
26	CA-2012-122490	2012-11-13	Ted Trueman	Furniture	Global Tron Executive Leather Low-Back Tiltar...	2404.7	349.83	150.26	

Query executed successfully.

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OBSERVATION:

Technology orders that exceed the category average are mostly

Copiers and Phones — the two highest-ticket sub-categories.

Furniture orders exceeding the category average are Bookcases

and Tables, but many still have negative profit.

-- D3. INNER JOIN – Customer's Orders vs Their Own Average

-- PURPOSE : Find customers with unusually large orders

SELECT

s.Order_ID,

s.Order_Date,

s.Customer_Name,

s.Segment,

s.Region,

ROUND(s.Sales, 2) AS Order_Sales,

ROUND(cust.Cust_Avg, 2) AS Customer_Avg_Sales

FROM dbo.Superstore AS s

INNER JOIN (

SELECT Customer_ID, AVG(Sales) AS Cust_Avg

FROM dbo.Superstore

GROUP BY Customer_ID

) AS cust ON s.Customer_ID = cust.Customer_ID

WHERE s.Sales > cust.Cust_Avg * 2 -- orders 2× the customer's own average

ORDER BY s.Sales DESC;

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	Order_ID	Order_Date	Customer_Name	Segment	Region	Order_Sales	Customer_Avg_Sales
1	CA-2014-145317	2014-03-18	Sean Miller	Home Office	South	22638.48	1669.54
2	CA-2016-118689	2016-10-02	Tamara Chand	Corporate	Central	17499.95	1587.68
3	CA-2017-140151	2017-03-23	Raymond Buch	Consumer	West	13999.96	839.85
4	CA-2017-127180	2017-10-22	Tom Ashbrook	Home Office	East	11199.97	1459.56
5	CA-2017-166709	2017-11-17	Hunter Lopez	Consumer	East	10499.97	1170.3
6	CA-2016-117121	2016-12-17	Adrian Barton	Consumer	Central	9892.74	723.68
7	CA-2014-116904	2014-09-23	Sanjit Chand	Consumer	Central	9449.95	642.83
8	US-2016-107440	2016-04-16	Bill Shonely	Corporate	East	9099.93	1166.85
9	CA-2016-158841	2016-02-02	Sanjit Engle	Consumer	South	8749.95	642.6
10	CA-2016-143714	2016-05-23	Christopher Conant	Consumer	East	8399.98	1102.64
11	CA-2014-143917	2014-07-25	Ken Lonsdale	Consumer	West	8187.65	488.8
12	CA-2014-139892	2014-09-08	Becky Martin	Consumer	Central	8159.95	736.85
13	US-2017-168116	2017-11-04	Grant Thornton	Corporate	South	7999.98	1558.54
14	CA-2014-145541	2014-12-14	Tom Boeckenhau...	Consumer	East	6999.96	537.29
15	CA-2015-145352	2015-03-16	Christopher Marti...	Consumer	South	6354.95	895.4
16	CA-2017-138289	2017-01-16	Andy Reiter	Consumer	Central	5443.96	734.27
17	US-2016-140158	2016-10-04	Daniel Raglin	Home Office	East	5399.91	642.37
18	CA-2017-143112	2017-10-05	Todd Sumrall	Corporate	East	5199.96	792.78
19	CA-2017-135909	2017-10-13	Jane Waco	Corporate	West	5083.96	551.55
20	CA-2016-136301	2016-03-13	Edward Hooks	Corporate	West	4912.59	322.21
21	US-2016-143819	2016-03-01	Karen Daniels	Consumer	East	4899.93	517.65
22	US-2015-128587	2015-12-24	Harry Marie	Corporate	Central	4899.93	411.84
23	CA-2017-149881	2017-04-01	Nick Crebassa	Corporate	West	4799.98	549.45
24	CA-2017-158379	2017-09-22	Joseph Airdo	Consumer	East	4663.74	270.46

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OBSERVATION:

These are "spike" orders — unusual one-time large purchases.

Useful for detecting bulk B2B orders or anomalous transactions.

-- D4. INNER JOIN – Region × Sub-Category Performance Matrix

-- PURPOSE : Cross-dimensional analysis of region and product

SELECT

s.Region,

s.Sub_Category,

rs.Total_Region_Sales,

ROUND(SUM(s.Sales), 2) AS SubCat_Sales_In_Region,

ROUND(SUM(s.Profit), 2) AS SubCat_Profit_In_Region,

ROUND(SUM(s.Sales) * 100.0

/ NULLIF(rs.Total_Region_Sales, 0), 2) AS Pct_Of_Region_Sales

FROM dbo.Superstore AS s

INNER JOIN (

SELECT Region, SUM(Sales) AS Total_Region_Sales

FROM dbo.Superstore

GROUP BY Region

) AS rs ON s.Region = rs.Region

GROUP BY s.Region, s.Sub_Category, rs.Total_Region_Sales

ORDER BY s.Region, SubCat_Sales_In_Region DESC;

100 % No issues found Ln: 138, Ch: 1 (597 chars, 16 lines) SPC CRLF UTF-8

Region	Sub_Category	Total_Region_Sales	SubCat_Sales_In_Region	SubCat_Profit_In_Region	Pct_Of_Region_Sales
Central	Chairs	501239.889593184	85230.65	6592.72	17
Central	Phones	501239.889593184	72403.28	12323.03	14.44
Central	Binders	501239.889593184	56923.28	-1043.64	11.36
Central	Storage	501239.889593184	45930.11	1969.84	9.16
Central	Tables	501239.889593184	39154.97	-3559.65	7.81
Central	Copiers	501239.889593184	37259.57	15608.84	7.43
Central	Accessories	501239.889593184	32956.08	7251.63	6.77
Central	Machines	501239.889593184	26797.38	-1486.07	5.35
Central	Bookcases	501239.889593184	24157.18	-1997.9	4.82
Central	Appliances	501239.889593184	23582.03	-2638.62	4.7
Central	Paper	501239.889593184	17491.9	6971.9	3.49
Central	Furnishings	501239.889593184	15254.37	-3906.22	3.04
Central	Supplies	501239.889593184	9467.37	-661.89	1.89
Central	Art	501239.889593184	5765.34	1195.16	1.15
Central	Envelopes	501239.889593184	4636.87	1777.53	0.93
Central	Labels	501239.889593184	2451.47	1073.08	0.49
Central	Fasteners	501239.889593184	778.03	236.62	0.16
East	Phones	678781.237776518	100614.98	12314.69	14.82
East	Chairs	678781.237776518	96260.68	9357.77	14.18
East	Storage	678781.237776518	71612.58	8389.37	10.55
East	Machines	678781.237776518	66106.16	6928.64	9.74
East	Binders	678781.237776518	53498	11267.93	7.88
East	Copiers	678781.237776518	53219.46	17022.84	7.84
East	Accessories	678781.237776518	45033.37	11195.86	6.63
East	Bookcases	678781.237776518	43819.33	-1167.63	6.46
East	Tables	678781.237776518	39139.81	-11025.38	5.77
East	Appliances	678781.237776518	34188.47	8391.41	5.04
East	Furnishings	678781.237776518	29071.38	5881.41	4.28
East	Paper	678781.237776518	20172.6	9015.37	2.97
East	Supplies	678781.237776518	10760.12	-1155.14	1.59

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OBSERVATION:

Phones and Chairs are top sub-categories in every region.

Tables consistently underperform profit-wise across ALL regions.

The Central region has the lowest % contribution from

high-margin Technology sub-categories.

-- D5. LEFT JOIN Simulation – Customers with Below-Average Profit

-- PURPOSE : Find customers contributing less than average

-- Step 1: Identify all customers and their total profit

-- Step 2: LEFT JOIN to overall average to flag underperformers

SELECT

cust.Customer_ID,

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cust.Customer_Name,

cust.Segment,

cust.Region,

ROUND(cust.Total_Sales, 2) AS Total_Sales,

ROUND(cust.Total_Profit, 2) AS Total_Profit,

ROUND(avg_profit.Avg_Profit, 2) AS Company_Avg_Profit,

CASE

    WHEN cust.Total_Profit < 0

    THEN 'Loss-Generating Customer'

    WHEN cust.Total_Profit < avg_profit.Avg_Profit

    THEN 'Below Average Profit'

    ELSE 'Above Average Profit'

END AS Customer_Profit_Status

FROM (

    SELECT Customer_ID, Customer_Name, Segment, Region,

           SUM(Sales) AS Total_Sales,

           SUM(Profit) AS Total_Profit

    FROM   dbo.Superstore

    GROUP BY Customer_ID, Customer_Name, Segment, Region

) AS cust

LEFT JOIN (

    SELECT AVG(cust_inner.Total_Profit) AS Avg_Profit

    FROM (

        SELECT Customer_ID, SUM(Profit) AS Total_Profit

        FROM   dbo.Superstore

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GROUP BY Customer_ID

) AS cust_inner

) AS avg_profit ON 1 = 1

ORDER BY cust.Total_Profit ASC;

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Results Messages

	Customer_ID	Customer_Name	Segment	Region	Total_Sales	Total_Profit	Company_Avg_Profit	Customer_Profit_Status
1	CS-12905	Cindy Stewart	Consumer	East	5016.55	-6892.37	361.69	Loss-Generating Customer
2	GT-14635	Grant Thornton	Corporate	South	8167.42	-3825.34	361.69	Loss-Generating Customer
3	LF-17185	Luke Foster	Consumer	Central	2680.69	-3805.55	361.69	Loss-Generating Customer
4	SR-20425	Sharelle Roach	Home Of.	West	2613.31	-3424.35	361.69	Loss-Generating Customer
5	HO-14965	Henry Goldwyn	Corporate	Central	1890.87	-2930.89	361.69	Loss-Generating Customer
6	NC-18415	Nathan Cano	Consumer	Central	1747.23	-2332.52	361.69	Loss-Generating Customer
7	NF-18385	Natalie Fritzier	Consumer	East	4863.55	-2057.33	361.69	Loss-Generating Customer
8	BM-11140	Becky Martin	Consumer	Central	10539.9	-1878.79	361.69	Loss-Generating Customer
9	JH-15985	Joseph Holt	Consumer	South	4297.64	-1862.31	361.69	Loss-Generating Customer
10	SM-20320	Sean Miller	Home Of.	South	23669.2	-1787.04	361.69	Loss-Generating Customer
11	TB-21520	Tracy Blumstein	Consumer	East	4087.14	-1720.71	361.69	Loss-Generating Customer
12	SB-20290	Sean Braxton	Corporate	East	1368.12	-1478.56	361.69	Loss-Generating Customer
13	ZC-21910	Zuschuss Carroll	Consumer	Central	3869.81	-1468.58	361.69	Loss-Generating Customer
14	KD-16270	Karen Daniels	Consumer	Central	1639.5	-1340.5	361.69	Loss-Generating Customer
15	TH-21550	Tracy Hopkins	Home Of.	South	1889.16	-1274.55	361.69	Loss-Generating Customer
16	DB-13120	David Bremer	Corporate	Central	760.98	-1141.47	361.69	Loss-Generating Customer
17	PF-19120	Peter Fuller	Consumer	Central	847.2	-1133.29	361.69	Loss-Generating Customer
18	DC-12850	Dan Campbell	Consumer	South	2447.51	-1121.99	361.69	Loss-Generating Customer
19	RO-19780	Rose O'Brian	Consumer	South	1998.84	-1097.86	361.69	Loss-Generating Customer
20	VW-21775	Victoria Wilson	Corporate	East	3725.51	-1078.13	361.69	Loss-Generating Customer
21	CP-12340	Christine Phan	Corporate	South	1475.2	-1065.75	361.69	Loss-Generating Customer
22	Dp-13240	Dean percer	Home Of.	South	3464.88	-1016.66	361.69	Loss-Generating Customer
23	JA-15970	Joseph Airdo	Consumer	East	4842.74	-1007.8	361.69	Loss-Generating Customer

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OBSERVATION:

Some customers have placed many orders yet generate NET LOSSES.

These are often customers who consistently request high discounts.

LEFT JOIN ensures ALL customers appear even if they don't match

a filter — important for complete auditing.

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-- D6. INNER JOIN – Year-over-Year Order Comparison (Self-Join)

-- PURPOSE : Compare a customer's orders across two years

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SELECT

y1.Customer_Name,

y1.Segment,

ROUND(y1.Sales_2016, 2) AS Sales_2016,

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ROUND(y2.Sales_2017, 2) AS Sales_2017,

ROUND(y2.Sales_2017 - y1.Sales_2016, 2) AS YoY_Growth

FROM (

SELECT Customer_ID, Customer_Name, Segment, SUM(Sales) AS Sales_2016

FROM dbo.Superstore

WHERE YEAR(Order_Date) = 2016

GROUP BY Customer_ID, Customer_Name, Segment

) AS y1

INNER JOIN (

SELECT Customer_ID, SUM(Sales) AS Sales_2017

FROM dbo.Superstore

WHERE YEAR(Order_Date) = 2017

GROUP BY Customer_ID

) AS y2 ON y1.Customer_ID = y2.Customer_ID

ORDER BY YoY_Growth DESC;

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Results Messages

	Customer_Name	Segment	Sales_2016	Sales_2017	YoY_Growth
1	Tom Ashbrook	Home Office	84.27	13723.5	13639.23
2	Raymond Buch	Consumer	914.06	14203.28	13289.22
3	Hunter Lopez	Consumer	456.56	10522.55	10065.99
4	Seth Vernon	Consumer	1041.21	8459.94	7418.72
5	Helen Wasser...	Corporate	1091.58	8166.35	7074.77
6	Andy Reiter	Consumer	27.24	5820.58	5793.34
7	Nick Crebassa	Corporate	58.42	5773.14	5714.73
8	Brian Moss	Corporate	2.91	5683.37	5680.47
9	John Lee	Consumer	7.71	5301.62	5293.91
10	Shahid Collister	Consumer	15.62	5259.49	5243.86
11	Tom Prescott	Consumer	179.54	4826.88	4647.34
12	Joseph Airdo	Consumer	477.5	5054.21	4576.71
13	Cathy Prescott	Corporate	243.96	4743.26	4499.29
14	Patrick O'Brill	Consumer	1272.87	5308.11	4035.24
15	Bill Epiett	Home Office	102.92	4101.76	3998.84
16	Pete Kriz	Consumer	2001.92	5979.14	3977.22
17	Jim Epp	Corporate	222.38	4074.03	3851.64
18	Ted Trevino	Consumer	5.98	3813.17	3807.19
19	Penelope Sewall	Home Office	12.9	3760.92	3748.02
20	Dean percer	Home Office	311.97	4032.76	3720.79
21	Robert Waldorf	Consumer	3.28	3492.36	3489.08
22	Dan Reichenba...	Corporate	700.28	4131.52	3431.24
23	Bart Watters	Corporate	394.2	3818.53	3424.33
24	Justin Deggeller	Corporate	1167.65	4508	3340.35
25	Jane Waco	Corporate	2056.01	5384.55	3328.54

Query executed successfully. localhost\SQLEXPRESS (17.0 ... | ARATI\HomePC (52) | CelebaI_Week2 | 00:00:00 | Row: 1, Col: 1 | 558 rows

OBSERVATION:

INNER JOIN here returns only customers who placed orders in

BOTH 2016 AND 2017 — i.e., returning customers.

Customers showing positive YoY growth are loyal, growing accounts.

Customers with negative growth might be churning.

-- D7. LEFT JOIN – Customers who ordered in 2016 but NOT 2017

-- PURPOSE : Identify churned customers

SELECT

prev.Customer_ID,

prev.Customer_Name,

prev.Segment,

ROUND(prev.Sales_2016, 2) AS Sales_2016,

curr.Sales_2017

FROM (

SELECT Customer_ID, Customer_Name, Segment, SUM(Sales) AS Sales_2016

FROM dbo.Superstore

WHERE YEAR(Order_Date) = 2016

GROUP BY Customer_ID, Customer_Name, Segment

) AS prev

LEFT JOIN (

SELECT Customer_ID, SUM(Sales) AS Sales_2017

FROM dbo.Superstore

WHERE YEAR(Order_Date) = 2017

GROUP BY Customer_ID

) AS curr ON prev.Customer_ID = curr.Customer_ID

WHERE curr.Customer_ID IS NULL -- ordered in 2016 but NOT in 2017

ORDER BY prev.Sales_2016 DESC;

	Customer_ID	Customer_Name	Segment	Sales_2016	Sales_2017
1	TC-20980	Tamara Chand	Corporate	18344.05	NULL
2	BS-11365	Bill Shively	Corporate	9199.79	NULL
3	KD-16270	Karen Daniels	Consumer	6317.16	NULL
4	JD-15895	Jonathan Doherty	Corporate	6181.11	NULL
5	RH-19510	Rick Huthwaite	Home Office	5973.97	NULL
6	JH-10690	Anna Haberlin	Corporate	5764.77	NULL
7	MC-17425	Mark Cousins	Corporate	3581.76	NULL
8	IM-15095	Ionia McGrath	Consumer	2832.55	NULL
9	RF-19345	Randy Ferguson	Corporate	2695.72	NULL
10	CM-12235	Chris Muller	Consumer	2395.71	NULL
11	DF-13135	David Flashing	Consumer	2067.8	NULL
12	JS-15880	John Stevenson	Consumer	1977.94	NULL
13	KT-16495	Kean Takahito	Consumer	1626.35	NULL
14	CA-12055	Cathy Armstrong	Home Office	1589.86	NULL
15	LH-17155	Logan Haushalt	Consumer	1503.24	NULL
16	DJ-13420	Denny Joy	Corporate	1454.9	NULL
17	MC-18130	Mike Caudle	Corporate	1444.04	NULL
18	PK-18910	Paul Knutson	Home Office	1422.25	NULL
19	KC-16255	Karen Carlisle	Corporate	1401.86	NULL
20	TB-21280	Toby Braunhardt	Consumer	1396.94	NULL
21	SP-20850	Stephanie Phelps	Corporate	1348.66	NULL
22	BS-11665	Brian Stugart	Consumer	1328.13	NULL
23	GH-14485	Gene Hale	Corporate	1288.46	NULL
24	ER-13955	Elpida Rittenbach	Corporate	1184.8	NULL
25	CD-12790	Cynthia Delaney	Home Office	1131.6	NULL
26	CH-12070	Cathy Hwang	Home Office	1127.98	NULL
27	IM-15070	Irene Maddox	Consumer	1108.98	NULL
28	JH-16180	Justin Hirsch	Consumer	1083.1	NULL
29	PM-18940	Paul MacInyre	Consumer	1056.86	NULL
30	SM-20905	Susan MacKen...	Consumer	1043.04	NULL
31	TS-21160	Theresa Swint	Corporate	1025.61	NULL
32	RC-19825	Roy Collins	Consumer	941.85	NULL

OBSERVATION:

LEFT JOIN is key here — it keeps ALL 2016 customers and identifies those with NO matching 2017 orders (NULL values).

These are potentially churned customers and prime targets for re-engagement campaigns.

Some high-value 2016 customers are not seen in 2017 —

a significant revenue risk.

-- D8. INNER JOIN – Segment × Region cross analysis

-- PURPOSE : Full cross-dimensional business intelligence view

SELECT

```

s.Segment,

s.Region,

ROUND(SUM(s.Sales), 2)          AS Total_Sales,

ROUND(SUM(s.Profit), 2)        AS Total_Profit,

COUNT(DISTINCT s.Order_ID)    AS Total_Orders,

ROUND(SUM(s.Sales) / COUNT(DISTINCT s.Order_ID), 2) AS Avg_Order_Value,

ROUND(SUM(s.Profit) * 100.0

      / NULLIF(SUM(s.Sales), 0), 2)  AS Profit_Margin_Pct

FROM dbo.Superstore AS s

INNER JOIN (

      SELECT DISTINCT Segment, Region

      FROM   dbo.Superstore

) AS seg_reg ON s.Segment = seg_reg.Segment

      AND s.Region = seg_reg.Region

GROUP BY s.Segment, s.Region

ORDER BY s.Segment, Total_Sales DESC;

```

100 % ✓ No issues found

	Segment	Region	Total_Sales	Total_Profit	Total_Orders	Avg_Order_Value	Profit_Margin_Pct
1	Consumer	West	362880.77	57450.6	825	439.86	15.83
2	Consumer	East	350908.17	41190.98	713	492.16	11.74
3	Consumer	Central	252031.43	8564.05	604	417.27	3.4
4	Consumer	South	195580.97	26913.57	444	440.5	13.76
5	Corporate	West	225855.27	34437.43	485	465.68	15.25
6	Corporate	East	200409.35	23622.58	434	461.77	11.79
7	Corporate	Central	157995.81	18703.9	348	454.01	11.84
8	Corporate	South	121885.93	15635.22	247	493.47	12.83
9	Home Office	West	136721.78	16530.41	301	454.23	12.09
10	Home Office	East	127463.73	26709.22	254	501.83	20.95
11	Home Office	Central	91212.64	12438.41	223	409.03	13.64
12	Home Office	South	74255	4620.63	131	566.83	6.22

OBSERVATION:

Best combination : Corporate + West → highest profit margin
Worst combination : Consumer + Central → negative margins possible
Home Office + East shows surprisingly strong average order values.